

University of Colombo School of Computing (UCSC)

University of Colombo
SCS 1307 – Probability and Statistics
Assignment 02 – Semester I – 2026
Answer Script

Question 1

Given $X \sim \text{Binomial}(10, p)$ and mean = 6.

(a) Find the probability of success

For a Binomial distribution, mean = np .

$$np = 6 \implies 10p = 6 \implies p = 0.6$$

Probability of success, $p = 0.6$

(b) Probability that exactly 7 successes occur

$$P(X = 7) = \binom{10}{7} p^7 (1-p)^3$$

$$= \binom{10}{7} (0.6)^7 (0.4)^3$$

$$\binom{10}{7} = 120, \quad (0.6)^7 = 0.0279936, \quad (0.4)^3 = 0.064$$

$$P(X = 7) = 120 \times 0.0279936 \times 0.064 = 0.21499$$

$$P(X = 7) \approx 0.2150$$

Question 2

A database server experiences an average of 2 failures per day, following a Poisson distribution, so $\lambda = 2$ per day. Failures on different days are independent.

The probability of at least one failure on a given day:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda} = 1 - e^{-2}$$

$$e^{-2} = 0.135335$$

$$P(X \geq 1) = 1 - 0.135335 = 0.864665$$

Since the two days are independent, the probability that at least one failure occurs on *each* of the next two days:

$$\begin{aligned} &P(\text{at least one failure on Day 1}) \times P(\text{at least one failure on Day 2}) \\ &= (0.864665)^2 = 0.74765 \end{aligned}$$

Required probability ≈ 0.7476

Question 3

$X \sim N(\mu = 120, \sigma = 15)$ (download speed in Mbps).

We need to find k such that:

$$P(120 - k < X < 120 + k) = 0.90$$

This is a symmetric interval about the mean, so standardizing:

$$P\left(-\frac{k}{15} < Z < \frac{k}{15}\right) = 0.90$$

Let $z = k/15$. Then:

$$2\Phi(z) - 1 = 0.90 \implies \Phi(z) = 0.95$$

From standard normal tables:

$$z = 1.645$$

Therefore:

$$k = 15 \times 1.645 = 24.675$$

$k \approx 24.675$ Mbps

Question 4

A cloud server receives an average of 100 API requests per minute, so $X \sim \text{Poisson}(\lambda = 100)$.

Using the normal approximation: $X \approx N(\mu = 100, \sigma^2 = 100)$, so $\sigma = 10$.

We require $P(90 \leq X \leq 110)$. Applying the continuity correction:

$$P(90 \leq X \leq 110) \approx P(89.5 < X < 110.5)$$

Standardizing:

$$z_1 = \frac{89.5 - 100}{10} = -1.05, \quad z_2 = \frac{110.5 - 100}{10} = 1.05$$

$$P(-1.05 < Z < 1.05) = \Phi(1.05) - \Phi(-1.05) = 2\Phi(1.05) - 1$$

From tables, $\Phi(1.05) = 0.8531$.

$$P = 2(0.8531) - 1 = 0.7062$$

Required probability ≈ 0.7062

Question 5

$X \sim \text{Binomial}(n = 200, p = 0.95)$.

Mean and variance:

$$\mu = np = 200 \times 0.95 = 190$$

$$\sigma^2 = np(1 - p) = 200 \times 0.95 \times 0.05 = 9.5, \quad \sigma = \sqrt{9.5} = 3.0822$$

We require $P(180 \leq X \leq 195)$. Applying the continuity correction:

$$P(180 \leq X \leq 195) \approx P(179.5 < X < 195.5)$$

Standardizing:

$$z_1 = \frac{179.5 - 190}{3.0822} = -3.407, \quad z_2 = \frac{195.5 - 190}{3.0822} = 1.7845$$

From standard normal tables:

$$\Phi(1.7845) \approx 0.9628, \quad \Phi(-3.407) \approx 0.0003$$

$$P = 0.9628 - 0.0003 = 0.9625$$

Required probability ≈ 0.9625

Question 6

$X \sim N(\mu = 250, \sigma = 40)$ (response time in milliseconds).

(a) $P(X < 300)$

$$z = \frac{300 - 250}{40} = 1.25$$

$$P(X < 300) = \Phi(1.25) = 0.8944$$

$$P(X < 300) \approx 0.8944$$

(b) $P(220 < X < 280)$

$$z_1 = \frac{220 - 250}{40} = -0.75, \quad z_2 = \frac{280 - 250}{40} = 0.75$$

$$P(220 < X < 280) = \Phi(0.75) - \Phi(-0.75) = 2\Phi(0.75) - 1$$

$$\Phi(0.75) = 0.7734$$

$$P = 2(0.7734) - 1 = 0.5468$$

$$P(220 < X < 280) \approx 0.5468$$

Question 7

A web server receives requests at an average rate of 6 per minute, so $X \sim \text{Poisson}(\lambda = 6)$.

We require $P(X < 4) = P(X \leq 3)$:

$$P(X \leq 3) = \sum_{k=0}^3 \frac{e^{-6} 6^k}{k!}$$

Calculating each term ($e^{-6} = 0.0024788$):

$$P(X = 0) = e^{-6} = 0.0024788$$

$$P(X = 1) = 6e^{-6} = 0.0148725$$

$$P(X = 2) = \frac{36}{2}e^{-6} = 18e^{-6} = 0.0446175$$

$$P(X = 3) = \frac{216}{6}e^{-6} = 36e^{-6} = 0.0892351$$

Adding these:

$$P(X \leq 3) = 0.0024788 + 0.0148725 + 0.0446175 + 0.0892351 = 0.1512039$$

Required probability ≈ 0.1512

End of Answer Script